Gaia Grosso

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Research interests. Statistical methods for detecting when data deviate from a reference model in high-dimensional settings: two-sample and goodness-of-fit testing with calibrated error control, interpretable kernel and physics-inspired models, and hypothesis testing in learned representation spaces. Applications to scientific discovery in particle physics and to auditing, monitoring and interpreting large AI models.

Academic Appointments

Sep 2026 – present	**Postdoctoral Associate**, Laboratory for Nuclear Science, Massachusetts Institute of Technology
Sep 2023 – Aug 2026	**IAIFI Postdoctoral Fellow**, NSF AI Institute for Artificial Intelligence and Fundamental Interactions *Laboratory for Nuclear Science, MIT · School of Engineering and Applied Sciences, Harvard University* Mentors: Prof. Philip Harris (MIT), Prof. Demba Ba (Harvard).
Nov 2019 – Jan 2023	**CERN Doctoral Student**, CMS Experiment, CERN, Geneva, Switzerland ML-based signal-agnostic searches with systematic uncertainties; graph neural networks for fast jet reconstruction on GPUs; monitoring and control software for the CMS 40 MHz Level-1 scouting system. Supervisor: Dr. Maurizio Pierini.
Jun – Sep 2018	**CERN Summer Student**, CERN, Geneva, Switzerland Machine learning techniques to search for new physics (report). Supervisors: Dr. Maurizio Pierini, Dr. Olmo Cerri.

Education

2019 – 2023	**Ph.D. in Physics**, University of Padova, Italy (research carried out at CERN) Thesis: *Searching for unexpected new physics at the LHC with machine learning*. Developed the New Physics Learning Machine (NPLM) and deployed it on CMS data and detector monitoring; results in [J2–J5]. Advisors: Prof. Marco Zanetti, Prof. Andrea Wulzer.
2017 – 2019	**M.Sc. in Physics**, University of Padova, Italy · 110/110 *cum laude* Thesis: *Unbiased ML-based strategy for new physics searches at the LHC*; results in [J1]. Advisor: Prof. Marco Zanetti.
2014 – 2017	**B.Sc. in Physics**, University of Padova, Italy Thesis: *Deep learning techniques to search for new physics at the LHC*. Advisor: Prof. Marco Zanetti.

Honors, Fellowships and Awards

2025	**Breakthrough Prize in Fundamental Physics**, awarded to the ATLAS, CMS, ALICE and LHCb Collaborations; co-recipient as a member of CMS.
2023 – 2026	**IAIFI Fellowship**, NSF AI Institute for Artificial Intelligence and Fundamental Interactions. Competitively selected independent postdoctoral fellowship.
2019 – 2023	**CERN Doctoral Student Programme**. Competitively selected doctoral fellowship at CERN.
2018	**CERN Summer Student Programme**. Competitively selected.

Publications

Bold: my name. † Student I mentored. * Equal contribution. $^{\alpha}$ Authors in alphabetical order.

Peer-reviewed conference papers and proceedings (2)

[C2] S. Bright-Thonney, C. Reissel, **G. Grosso**, N. Woodward†, E. Govorkova, A. Novak, S.E. Park, E. Moreno, P. Harris. AutoSciDACT: Automated scientific discovery through contrastive embedding and hypothesis testing. *Advances in Neural Information Processing Systems* **38** (NeurIPS 2025), 88575–88614. arXiv:2510.21935

[C1] **G. Grosso**, N. Lai, M. Migliorini, J. Pazzini, A. Triossi, M. Zanetti, A. Zucchetta$^{\alpha}$. Triggerless data acquisition pipeline for machine learning based statistical anomaly detection. *EPJ Web Conf.* **295**, 02033 (2024). Proceedings of CHEP 2023.

Preprints and papers under review (5)

[P5] S.X. Yu†, **G. Grosso**. When harm looks normal: auditing fine-tuned models via representation drift. Under review (AISTATS 2027).

[P4] C. McNamee†, H. Orgad*, **G. Grosso***, N. Saphra, D. Ba. To be or not to be: mutually exclusive features in language models. Under review (ICLR 2027).

[P3] S. Tong†, P. Harris, **G. Grosso**. Covariant contrastive learning for uncertainty-aware anomaly detection. arXiv:2609.35755 (2026).

[P2] **G. Grosso**, S.S.R. Hindupur†, T. Fel, S. Bright-Thonney, P. Harris, D. Ba. Sparse, self-organizing ensembles of local kernels discover rare statistical anomalies. arXiv:2511.03095 (2025). Under review at *PNAS*.

[P1] N.S. Woodward†, S.E. Park, **G. Grosso**, J. Krupa, P. Harris. Product manifold machine learning for physics. arXiv:2412.07033 (2024).

Peer-reviewed journal articles (10)

[J10] P. Cappelli†, **G. Grosso**, M. Letizia, H. Reyes-Gonzalez, M. Zanetti. Learning to validate generative models: a goodness-of-fit approach. *Mach. Learn.: Sci. Technol.* **7**, 045011 (2026). arXiv:2511.09118

[J9] K. Metzger†, M. Sodini, L. Xu†, T. Aarrestad, E. Govorkova, **G. Grosso**, P. Harris. Anomaly-preserving contrastive neural embeddings for end-to-end model-independent searches at the LHC. *Phys. Rev. D* **112**, 072011 (2025).

[J8] **G. Grosso**, D. Sengupta, T. Golling, P. Harris. Robust resonant anomaly detection with NPLM. *Eur. Phys. J. C* **85** (2025).

[J7] **G. Grosso***, M. Letizia*. Multiple testing for signal-agnostic searches for new physics with machine learning. *Eur. Phys. J. C* **85**, 4 (2025). arXiv:2408.12296

[J6] **G. Grosso**. Anomaly-aware summary statistic from data batches. *J. High Energy Phys.* **12** (2024) 093. *Single author.* arXiv:2407.01249

[J5] **G. Grosso**, M. Letizia, A. Wulzer, M. Pierini. Goodness of fit by Neyman–Pearson testing. *SciPost Phys.* **16**, 123 (2024).

[J4] **G. Grosso**, N. Lai, M. Letizia, J. Pazzini, M. Rando, L. Rosasco, A. Wulzer, M. Zanetti$^{\alpha}$. Fast kernel methods for data quality monitoring as a goodness-of-fit test. *Mach. Learn.: Sci. Technol.* **4**, 035029 (2023).

[J3] M. Letizia, G. Losapio, M. Rando, **G. Grosso**, A. Wulzer, M. Pierini, M. Zanetti, L. Rosasco. Learning new physics efficiently with nonparametric methods. *Eur. Phys. J. C* **82**, 879 (2022).

[J2] R.T. D'Agnolo, **G. Grosso**, M. Pierini, A. Wulzer, M. Zanetti$^{\alpha}$. Learning new physics from an imperfect machine. *Eur. Phys. J. C* **82**, 275 (2022). arXiv:2111.13633

[J1] R.T. D'Agnolo, **G. Grosso**, M. Pierini, A. Wulzer, M. Zanetti$^{\alpha}$. Learning multivariate new physics. *Eur. Phys. J. C* **81**, 89 (2021).

Workshop papers (*non-archival*) (3)

[W3] **G. Grosso**, A. Marin-Llobet, D. Ba. Adaptive and interpretable two-sample test as a batch-level tool for AI safety. ICML 2026 Workshop on Mechanistic Interpretability, Seoul.

[W2] G. Badaracco†, S. Benevedes, **G. Grosso**, C. Reissel, T. Aarrestad, P. Harris. An end-to-end pipeline for uncertainty-aware validation of generative AI. NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences, San Diego.

[W1] **G. Grosso**, E. Govorkova, P. Harris, E. Moreno, R. Raikman. Learning dictionaries of new physics with sparse local kernels. NeurIPS 2024 Workshop on Machine Learning and the Physical Sciences, Vancouver.

Reviews, perspectives and white papers (3)

[R3] **G. Grosso**, R. Winterhalder, L. Brenner, L. Lyons, T. Plehn. VERAIPHY – Validation & Evaluation for Robust AI in PHYsics. Opening article of the PHYSTAT VERAIPHY series. arXiv:2608.17724 (2026).

[R2] **G. Grosso**, V. Mikuni, L. Heinrich. Are we ready for AI-driven discovery? AI verification before the next fundamental physics breakthrough. Review, VERAIPHY series. arXiv:2607.10039 (2026). To be submitted to *SciPost Physics*.

[R1] **G. Grosso**, P. Harris, S. Mishra-Sharma, P. Shanahan. A virtuous cycle: generative AI and discovery in the physical sciences. *An MIT Exploration of Generative AI* (2024).

Collaboration papers

Member of the CMS Collaboration since 2018; on the author list since 2021 and co-author of 465 CMS Collaboration papers (full list on INSPIRE-HEP).

Patents

2025	**G. Grosso**, D.E. Ba, S.S. Hindupur Ravindra†. Methods for explainable novelty discovery using artificial intelligence. U.S. Provisional Patent Application No. 63/909,442, filed October 31, 2025. Covers the SparKer method [P2].

Software

NPLM	New Physics Learning Machine: ML implementation of the likelihood-ratio test for signal-agnostic anomaly detection; calibrated p-values and localization of the most discrepant regions [J1–J5, J7, J8]. GitHub
SparKer	Sparse, self-organizing ensembles of local kernels for rare-anomaly detection in learned representations, with geometrically interpretable detection experts [P2, W3]. Code covered by a U.S. provisional patent application (see Patents).
Tutorials	Hands-on notebooks on diffusion models, VAEs and generative-model validation, written for the IAIFI Summer School 2024. GitHub

Invited Talks, Seminars and Panels

Sep 2026	*Discovery through the lens of AI: scaling anomaly detection with learned representations.* LPC Topic of the Week, LHC Physics Center, Fermilab, Batavia, IL. [event] [slides]
Apr 2026	*Detecting statistical anomalies with sparse, self-organizing ensembles of local kernels.* Statistical Physics for Ambitious Interpretability, Cambridge, MA. [event]
Oct 2025	*From high dimensions to statistical discovery: anomaly detection at scale in the era of latent spaces.* STAMPS–NSBI Workshop, Carnegie Mellon University, Pittsburgh, PA. [video]
Mar 2025	Panelist, round table on goodness-of-fit tests via machine learning. PHYSTAT Informal Review (remote). [event]
Sep 2024	*Statistical tests for anomaly detection at the LHC.* PHYSTAT – Statistics meets ML in Particle Physics & Astrophysics, Imperial College London, UK. [programme]
Nov 2023	*Uncertainty-aware ML models.* From Physics to Medicine: XAI Workshop. [event]
Sep 2022	Seminar: *New Physics Learning Machine (NPLM) at work.* Galileo Galilei Institute, Florence, Italy. [event]

Contributed Talks

Sep 2025	*Validation, monitoring and discovery: statistical tests for anomalies in the era of latent spaces.* FirstPrinciple Talks Program (remote).
Sep 2025	*Anomaly detection in the era of latent spaces.* Building Big or Building Smart, MIAPbP, TU Munich, Germany. [slides]

Jun 2025	*From high dimensions to statistical discovery: a contrastive learning approach to anomaly detection.* Anomaly Detection for HEP Workshop, Columbia University, Irvington, NY. [slides]
Aug 2024	*How good is your model? Goodness of fit by Neyman–Pearson testing.* IAIFI Summer Workshop, MIT, Cambridge, MA.
May 2024	*Boosting statistical anomaly detection via multiple testing with NPLM.* ACAT 2024, Stony Brook University, NY. [event]
Jun 2023	*Goodness of fit by Neyman–Pearson testing.* PHYSTAT-2sample (remote). [event]
May 2023	*Systematics in ML model-independent searches.* CHEP 2023, Norfolk, VA. [event]
Apr 2023	*Unbiased detection of data departures from expectations with machine learning.* Systematic Effects and Nuisance Parameters in Particle Physics Data Analyses, BIRS, Banff, Canada. [event]
Sep 2022	*Learning new physics from a machine.* Machine Learning at GGI, Florence, Italy. [event]
Aug 2022	*New Physics Learning Machine (NPLM) at work.* Hammers & Nails, Weizmann Institute of Science, Rehovot, Israel. [event]
May 2022	*Learning new physics aware of systematic uncertainties.* 5th Inter-experiment Machine Learning Workshop, CERN, Geneva. [event]
Apr 2022	*An imperfect machine to search for new physics: dealing with uncertainties in a machine-learning based signal extraction.* Learning to Discover, Institut Pascal, Paris-Saclay, France. [event]
Apr 2022	*Model-independent search for new physics at the LHC.* Spring Workshop on Physics of Data, Venice, Italy. [event]
Dec 2021	*An imperfect machine to search for new physics: a way to include systematic uncertainties.* ACAT 2021 (remote). [event]

Posters

Jul 2026	*Adaptive and interpretable two-sample test as a batch-level tool for AI safety* [W3]. ICML 2026 Mechanistic Interpretability Workshop, Seoul, South Korea.
Dec 2024	*Learning dictionaries of new physics with sparse local kernels* [W1]. NeurIPS 2024 ML4PS Workshop, Vancouver, Canada.
Mar 2024	*Building sparse kernel methods via dictionary learning: expressive, regularized and interpretable models for statistical anomaly detection.* EuCAIFCon 2024, Amsterdam, Netherlands. [event]
Dec 2022	*How good is the Standard Model? Machine learning multivariate goodness-of-fit tests.* NeurIPS 2022 ML4PS Workshop, New Orleans, LA. [paper]
Dec 2021	*An imperfect machine to search for new physics: systematic uncertainties in a machine-learning based signal extraction.* NeurIPS 2021 ML4PS Workshop (virtual).
	Co-authored posters presented by mentees and collaborators: Conference on Physics and AI (PAI 2026, Stanford) [P3]; NeurIPS 2025 main conference [C2] and ML4PS Workshop [W2]; EuCAIFCon 2025 (Cagliari); NeurIPS 2024 ML4PS Workshop [P1]; NeurIPS 2022 ML4PS Workshop.

Teaching

Guest lectures in university courses

Mar 2026	*Same or different? Classical and ML methods for two-sample testing in science.* CDS DS 595 AI Methods for Science, Boston University. [slides]
Apr 2025	*Statistical tests for anomaly detection and scientific discovery.* 8.16/8.316 Data Science in Physics, MIT.

Lectures and tutorials at schools

Feb 2027	Lecturer (upcoming), PHYSTAT School of Statistics, Damp, Germany (Feb 22 – Mar 1, 2027).
Nov 2025	Lecturer, *Statistical and ML tools for anomaly detection in fundamental physics.* First PHYSTAT School of Statistics: Statistics in the Era of Machine Learning, Nikhef, Netherlands.
Aug 2025	Lecturer, *Physics-guided optimizations: inform and inspire AI models with physics principles.* IAIFI Summer School, Harvard University. [slides]

Aug 2024 | Tutorial developer and facilitator, *Diffusion models for physics*: designed notebooks on diffusion models, VAEs and validation; led the hands-on session. IAIFI Summer School, MIT. [notebooks]

Mentoring

14 students mentored since 2023 (4 undergraduate, 10 graduate and Master's) at MIT, Harvard, Princeton, ETH Zurich, University of Padova, IFT Madrid and IFAE Barcelona; 9 papers, preprints and submissions co-authored with mentees (marked † in the publication list). Primary research advisor unless noted; MIT undergraduates were supervised through the Undergraduate Research Opportunities Program (UROP). Linked project titles lead to the resulting papers.

Period	Student	Project
Current		
May 2026 –	Shirley X. Yu, undergraduate, Princeton	Auditing fine-tuned language models via representation drift
Feb 2026 –	Francesco Montagno, graduate, IFAE Barcelona	Anomaly detection in the four-lepton final state with CMS Open Data; interpretable EFT searches with contrastive latent spaces
Sep 2025 –	Cameron McNamee, graduate, Harvard (co-advised)	Robust, interpretable local layers for supervised learning
May 2025 –	Sergio Rodriguez Benitez, graduate, IFT Madrid	Regularization and the look-elsewhere effect in ML-based anomaly detection; contributing to a review in preparation
Apr 2025 –	Shelley Tong, graduate, MIT	Continuous systematic uncertainties in contrastive embeddings for LHC data; anomaly detection in the four-lepton final state with CMS Open Data
Sep 2023 –	Nicolò Lai, graduate, Univ. of Padova	Signal-agnostic NPLM search in the CMS dimuon final state
Former		
Mar – Aug 2026	Ahmed Fadhlallah, undergraduate, MIT	Metrics for evaluating embedding spaces for anomaly detection in the four-lepton final state with CMS Open Data
Aug – Oct 2025	Sai Sumedh R. Hindupur, graduate, Harvard	Theory of SparKer learning dynamics
Nov 2024 – Sep 2025	Giada Badaracco, Master's, ETH Zurich (co-advised)	Uncertainty-aware validation of generative AI
Sep 2023 – Sep 2025	Pietro Cappelli, graduate, Univ. of Padova	Generative-model validation via ML goodness-of-fit tests
Oct 2023 – Apr 2025	Lana Xu, undergraduate, MIT	Contrastive representations for anomaly detection
Apr 2023 – Apr 2025	Nathaniel S. Woodward, undergraduate, MIT	Product-manifold machine learning for particle physics
Oct 2023 – Sep 2024	Kyle S. Metzger, Master's, ETH Zurich (co-advised)	Representation learning for new physics searches in CMS
Jan – Jun 2024	Jessica Prendi, Master's, ETH Zurich (co-advised)	Model-agnostic searches in the CMS 40 MHz scouting system

Leadership and Service

Leadership roles

2026 – present | **Co-organizer, PHYSTAT**: member of the event-organizer core team and of the Advisory Committee
International series of workshops, seminars and schools on statistical methods in particle physics, bringing together physicists and statisticians (founded 2000). [webpage]

2025 – present | **Co-organizer, VERAIPHY** (Validation & Evaluation for Robust AI in Physics), PHYSTAT-endorsed initiative
Article series on statistical standards for ML in fundamental physics [R2, R3]; workshop (Feb 2026). [webpage]

2025 – 2026 | **Track convener**, "Software environment and maintainability," CHEP 2026, Bangkok, Thailand (May 2026). Abstract selection, session chairing, proceedings review.

2023 – 2025	**Convener, ML Innovation (L3)**, CMS Machine Learning Group, CERN Organized seminars, workshops and hackathons on emerging ML methods for CMS. Hackathons (internal): ML for Reweighting (Oct 2023); Anomaly Detection for Monitoring, Validation and Discovery (Jul 2024). Public workshops: *Foundation Models for Physics* (Oct 2024) and the *Bites of Foundation Models for Physics* series (Nov 2024, Jun 2025).
2024 – 2025	**Co-chair, IAIFI Speaker Selection Committee**: seminar and colloquium series.

Event organization

Mar 2025	Co-organizer, IAIFI Research Panel on the 2024 Nobel Prize in Physics, MIT. [video]
Nov 2024	Co-organizer, IAIFI Fireside Chat on the 2024 Nobel Prize in Physics, MIT. [video]
Mar 2024	Organizing committee, IAIFI Symposium on the Impact of Generative AI in the Physical Sciences, MIT; co-authored the resulting white paper [R1].

Refereeing

Journals	*PRX Intelligence*, *Physical Review D*, *Nuclear Physics B*, *Journal of High Energy Physics*, *European Physical Journal C*.
Conferences	ICLR, AISTATS, NeurIPS Workshop on Machine Learning and the Physical Sciences.

Outreach

Nov 2024	Cambridge Science Festival: built an LLM-powered, retrieval-augmented interactive Marie Curie persona answering visitors' questions in historically grounded form.
Nov 2024	Museum of Science, Boston: public talk and Q&A on AI and particle physics.
Sep 2023	Public lecture, *How good is the Standard Model? Hunting anomalies in the LHC data and beyond.* Institute of Advanced Study, University of Geneva. [event]
2023	CERN Science Gateway: video recording on physics at the LHC for the permanent exhibition.
Nov 2020	Veneto Night: livestreamed presentation on new physics searches for the general public and children, University of Padova. [video]
2018	Built a cloud chamber with a student team for particle-detection demonstrations to high-school students, University of Padova.

Languages

Italian (native), English (fluent), Catalan (intermediate), French (basic).